

chain, from the OEM to the semiconductor vendor and IP provider. Expert collaborations from inside as well as outside the automotive electronics systems industry will be required to solve the security, user interface and system availability challenges for such autonomous cars.

Lack of purchasing power. EVs are again gaining popularity now, but not without any challenge. The main problem is the people's perspective for these vehicles and also the fact that these cost more than the normal cars. There seems to be no reason to switch from normal to electric cars even if they have benefits for the future. However, there are some design houses who have come up with new designs which can change the way the people traditionally think about EVs by creating a whole new different section for these cars.

Legal challenges. Simran Kaur, head of sales, Speedways Electric says, "As we look at the trend of the EVs in India, we find that we have almost no choice as there is either one or two models available which look like traditional cars. So what is that would make people switch from normal cars to electric cars, agreeing to the fact that the EVs cost more than the normal cars. So we are trying to change the way the people traditionally think about EVs."

But that again poses the legal challenges for such concepts. "The major issues with these type of vehicles is the approval to get them running on roads because there is no such category provided by the government, and the category provided is that of traditional cars where they do not have any concept of future vehicle," Simran adds.

Challenges faced by design houses. The major challenges in incorporating electronics in an automotive are the harsh and tough operational conditions. The automotive electronics faces much tougher operating environment as compared to consumer electronics, especially due to the vibrations and the temperature conditions such as electronics located close to the engines. Here you have to ensure that the electronic components are fully reliable

Auto industry needs support

Last year, the overall growth in the auto industry was not promising enough. Overall automobile exports grew by 6.08 per cent during April-December 2013. But if we look at the overall domestic sales last year, it grew by a margin of only 2.30 per cent because of growth in two-wheelers' sales with overall passenger vehicle segment having a negative growth of (-)5.72 per cent and commercial vehicles were also sidelined in the market with a de-growth of (-)18.38 per cent. While three-wheelers' sales declined by 9.31 per cent last year, the domestic sales were somehow upheld and supported by a positive growth of 5.40 per cent in the two-wheeler segment.

The auto industry went through a rough time last year registering an overall growth of only 2.15 per cent as compared to the previous year's statistics, and this growth was all due to the growth in two-wheelers section. The automakers are hopeful that the improvement in our economy and innovations in the field of automotive electronics together might turn the numbers for them.

and withstand high temperature and humidity conditions.

With more and more electronics incorporated in an automotive, the power capacity is increasing and therefore cooling has become one of the biggest challenge with automotive electronics. The device sizes are decreasing and the automobiles have to be lighter, so traditional cooling techniques are no longer useful; one needs to come up with new techniques.

Another important thing is the electromagnetic aspect because having a lot of electronic circuitry in the automobiles will involve electromagnetic interference as well as compatibility. Having a lot of high-frequency devices in cars for communication can cause interference between those devices that may lead to problems. Therefore it poses a challenge before the designers who need to have a clear electrical as well as electromagnetic modelling.

Measures taken by the government

Automotive electronics has evolved as electronics industry has made huge strides first. Electronics industry is growing in India but still most of the required electronics have to be imported, as there are no chip manufacturing companies in India. This increases the cost which is always an important factor in developing markets such as India.

The government of India has been encouraging and promoting investments in electronics systems design

and manufacture (ESDM). It has taken several initiatives for the development of electronics sector in the country. The government has recently approved National Policy on Electronics (NPE). The objective of the NPE is to achieve a turnover of about USD 400 billion by 2020 involving investment of about USD 100 billion and employment to around 28 million people.

It will include achieving a turnover of USD 55 billion of chip design and embedded software industry, USD 80 billion of exports in the sector. Moreover, the policy also proposes setting up of over 200 electronic manufacturing clusters. Another important objective of the policy is to significantly upscale high-end human resource creation to 2500 PhDs annually by 2020 in the sector.

An ever-growing industry

With all the drivers' assistance and safety systems and special utilities, automobiles are also becoming more smart and intelligent. Owing to the endless number of demands by the consumers, the industry is ever growing and the major growth is expected from the emerging economies such as India.

Being one of the key drivers of the automotive industry and the rising demands on innovation, solutions, cost, supply chain and services leading its way forward in the Indian market, we can see the automotive electronics shifting its gears in the automotive industry. ●

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